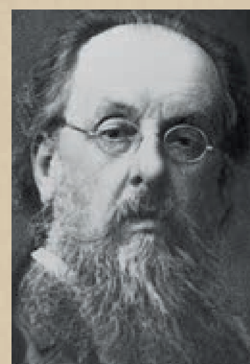

"First comes, inevitably, the idea, the fantasy, the fairy tale. Then comes scientific calculation. Ultimately, fulfillment crowns the dream."

**Konstantin Tsiolkovsky**
EARLY THEORIST OF ROCKET FLIGHT INTO SPACE, 1926

# JOURNEY TO THE MOON

## IN HALF A CENTURY MANNED SPACE TRAVEL GREW FROM THE OBSESSION OF A FEW DREAMERS INTO A DESPERATE RACE BETWEEN COLD WAR RIVALS

**In 1903, as the** Wright brothers were approaching success at Kitty Hawk, an unknown Russian schoolteacher, Konstantin Tsiolkovsky, published a paper entitled "The Exploration of Space with Reaction Propelled Devices" in a small-circulation scientific journal. Although the world at large paid no attention, Tsiolkovky had theoretically solved the basic problem of manned space flight – how to propel human beings out of the Earth's atmosphere without killing them.

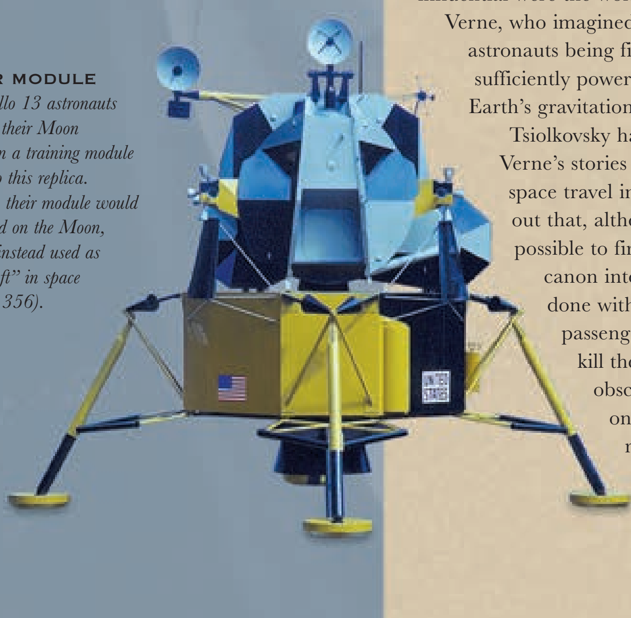


**RUSSIAN PIONEER**
*Konstantin Tsiolkovsky cracked many of the problems of space flight in theory, but not in practice.*

Fantasy about space flight had a long history. In the seventeenth century, when the Moon and planets had only recently been identified as bodies moving in space, astronomer Johannes Kepler fantasized about a journey to the Moon in one of the first science fiction stories, *Somnium*. By the nineteenth century, Moon travel had developed into a familiar element of fantasy fiction. Most influential were the works of French author Jules Verne, who imagined his Moon-bound astronauts being fired from a cannon sufficiently powerful to overcome the Earth's gravitational pull.

Tsiolkovsky had been inspired by Verne's stories to begin research into space travel in the 1880s. He figured out that, although it was theoretically possible to fire a projectile from a canon into space, it could not be done without subjecting its passengers to forces that would kill them. Working in provincial obscurity, Tsiolkovsky lighted on the correct solution: a rocket. Following Newton's third law of motion, that every action produces an equal and opposite reaction, it would operate as a propulsion system both in and outside the atmosphere. Tsiolkovsky also figured out what fuel would power the rocket engine, suggesting liquid hydrogen and liquid oxygen to provide the required thrust. In the early 1900s, he went on to design spacecraft with prescient refinements such as steering vanes and multistage launchers, as well as a winged proto-Shuttle capable of gliding back to Earth.

Tsiolkovsky's eccentric interest in space rockets was shared by a number of isolated individuals in different countries who, up to the 1920s, pursued the subject with little or no reference to one another's work. In France, aviation pioneer Robert Esnault-Pelterie gave lectures on the possibility of space travel in 1912. In the US, by 1909 physicist Robert Goddard, then a postgraduate student at Clark University, had calculated the velocities required to reach Earth orbit. He carried out research into solid-fuel rocket engines and in 1919 attracted press notice by stating that a rocket could reach the Moon. Goddard said nothing of manned space flight, but in 1923 German teacher Hermann Oberth published *Die Rakete zu den Planetenräumen* (The Rocket into Interplanetary Space), assessing the practicality of human space travel, and giving plans for a liquid-propellant two-stage rocket.



**LUNAR MODULE**
*The Apollo 13 astronauts practiced their Moon landing in a training module similar to this replica. However, their module would never land on the Moon, but was instead used as a "life raft" in space (see page 356).*